

A Strange Idea, Followed Further

Step 6.3, Actually Done, and What That Showed

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Abstract

The previous document stopped one section short of doing the calculation it described. This one does the calculation. The result is in every respect consistent with the standard result of Chamseddine–Connes–van Suijlekom (2013). Along the way, one parenthetical statement of the previous document was found to be wrong as written and is corrected here; one numerical claim was redone with current data; and the open question that the previous document called Step 4 is located one click more precisely. The strange idea did not become any less strange. It became better mapped.

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1 What this document is, and why it exists

The previous document¹ closed with Section 6.3, in which we had attempted to set up the explicit $(\hat{A}, J, \gamma)$ -bimodule for the Pati–Salam algebra in code, verified the KO-dimension 6 conditions, and then stopped. The stated reasons for stopping were:

- (1) Reproducing the published calculation adds no new information.
- (2) The genuine open question is not answered by reproducing the calculation.

Both reasons are defensible. Neither, on reflection, is sufficient. The calculation that was paraphrased in three lines is also the calculation that, if one wants to test *anything* downstream of the order-one carve-out, one needs to actually have running. The previous document had a parenthetical assertion about A_F failing to embed as a unital $*$ -subalgebra of the Pati–Salam algebra; that assertion is, under a literal reading, false, and it was made because we had not yet built the embedding. The previous document also described the Koide residual to seven decimal places using a number whose provenance it did not carefully chase; that number is also approximately right but is scheme-dependent, and the provenance turns out to matter slightly.

This document records the calculation done. It is, like its predecessor, not a theory and not a framework. It is a record. The audience is presumed to be familiar with the previous document; the section numbering continues from there.

2 Recap, in two paragraphs

The Generator hypothesis was a maximalist claim that the Standard Model and gravitation are consequences of a single self-referential algebraic object. Most of its alleged empirical content turned out to be fitted rather than derived; a structural rigidity theorem ruled out its central object under its own axioms; the honest rewrite reduced the framework to operator-algebraic renormalization group theory, which is real, well-studied, and predicts nothing new. The residual mathematical question, on which the framework’s chance of becoming non-trivial entirely rests, is the selection of the finite algebra $A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ in Connes’s spectral action program. This is Step 4. As of May 2026 nobody has closed it.

The previous document located Step 4 inside the algebra-enumeration step of Chamseddine–Connes (2007) and at the level of the bimodule decoration. It did not, however, build the bimodule. The present document does.

3 The Koide residual, redone carefully

Before the structural work, one minor numerical cleanup. The previous document quoted

$$K = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = 0.666661 \pm 0.000007,$$

deviation from $2/3$ at the 10^{-5} level. The provenance of the value was not specified, and the uncertainty was given without specifying the input set.

Recomputed in this session against current data:

¹V. Savytsky with Claude (Anthropic), *A Strange Idea, Followed Honestly: The Generator Hypothesis After Every Stress Test We Could Apply*, May 2026.

- With CODATA 2022 values for m_e , m_μ , m_τ (the last from PDG 2022 propagated into CODATA): $K = 0.6666605(1)$, deviation from $2/3$ of $\sim 9 \times 10^{-6}$, statistical distance from $2/3$ approximately 0.91σ under the quoted uncertainties.
- With PDG 2024 values directly, and in particular the updated $m_\tau = 1776.93 \pm 0.09$ MeV: $K = 0.6666645(5)$, deviation from $2/3$ of $\sim 2 \times 10^{-6}$, statistical distance approximately 0.43σ .

Observation 1. *For exact agreement $K = 2/3$, the tau mass would need to shift by between $+0.04$ and $+0.11$ MeV, well inside the current PDG uncertainty.*

Observation 2. *K is computed from pole masses. Under one-loop pure QED running it is RG-invariant; under full Standard Model running it is not. The “ 10^{-5} level coincidence” is therefore a coincidence in a specific renormalization scheme. Under a different scheme the deviation can be order 10^{-3} or larger.*

This does not refute the relation. It does mean that the previous document’s report of seven-decimal agreement was, on close inspection, agreement in a scheme that the document did not name. The honest version is: K is consistent with $2/3$ within current experimental uncertainty, in the pole-mass scheme; the deviation is at most 1σ , possibly less; no derivation of why this should hold exists in any rigorous framework. The relation does not extend cleanly to quarks. It may be coincidence; it may be a hint; it is currently undecidable. This is the same verdict the previous document reached, with one fewer false-precision figure.

4 The embedding, actually constructed

The previous document, in Observation 9, stated:

Of the four CC07-irreducible candidates at fermion dimension 32 ($M_4(\mathbb{C})$, $M_4(\mathbb{R}) \oplus M_4(\mathbb{R})$, $M_2(\mathbb{H}) \oplus M_4(\mathbb{R})$, $M_2(\mathbb{H}) \oplus M_2(\mathbb{H})$), none contains A_F as a unital $$ -subalgebra. Even the canonical Pati–Salam algebra $M_2(\mathbb{H}) \oplus M_4(\mathbb{C})$ does not.*

The parenthetical about $M_2(\mathbb{H}) \oplus M_4(\mathbb{C})$ is, taken at face value, wrong. An explicit unital $*$ -embedding exists and we constructed it.

Theorem 3. *There exists a unital $*$ -algebra homomorphism*

$$\iota: \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}) \longrightarrow M_2(\mathbb{H}) \oplus M_4(\mathbb{C})$$

with zero kernel.

Proof. Explicit construction. View $\mathbb{C} \subset \mathbb{H}$ as the subalgebra spanned by $\{1, i\}$. For $(\lambda, q, X) \in \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$, define

$$\iota(\lambda, q, X) = \left(\text{diag}(\lambda, q) \in M_2(\mathbb{H}), \text{diag}(\lambda, X) \in M_4(\mathbb{C}) \right),$$

where on the right $\text{diag}(\lambda, X)$ places λ in the $(1, 1)$ entry and X in the lower 3×3 block. Unitality, additivity, $\mathbb{R}$ -linearity, $*$ -preservation and faithfulness all follow by direct check. Multiplicativity reduces to multiplicativity of the inclusion $\mathbb{C} \hookrightarrow \mathbb{H}$ and of block-diagonal matrix multiplication. The image is unital because both blocks are. The kernel is zero because each summand of the source is faithfully represented in at least one summand of the target. $\square$

We verified this numerically as a sanity check on the embedding. Over fifty independent random elements of A_F :

- unitality residual: 0;
- multiplicativity residual: $\leq 1.78 \times 10^{-15}$ (machine epsilon, double precision);
- additivity, $\mathbb{R}$ -linearity, $*$ -preservation: residuals all 0;
- faithfulness check (nonzero elements mapping to zero): 0 out of 100.

This is not a deep correction. The previous document’s Observation 9 was attempting to express a real and well-known fact in the NCG literature, namely that A_F is not a *canonical multiplicity-1* unital $*$ -subalgebra of the Pati–Salam algebra. The K-rank obstruction it pointed to ($10 \neq 12$) is real. But the parenthetical sentence, as written, slid from “no canonical inclusion” to “no inclusion”, which is a different and stronger and false claim. The right statement is the one Chamseddine–Connes–van Suijlekom 2013 make: A_F is carved out of the Pati–Salam algebra at the level of the (J, γ) -decorated bimodule, by the order-one condition acting on a fixed grading, not by the existence or non-existence of an inclusion of algebras at the matrix level. Both relationships hold; one is much more informative than the other; the previous document was making a sharp point and overshooting slightly. The replacement statement is:

Observation 4. *A_F embeds as a unital $*$ -subalgebra of $M_2(\mathbb{H}) \oplus M_4(\mathbb{C})$ via a multiplicity-2 / corner-3 inclusion. It does not embed as a canonical multiplicity-1 unital $*$ -subalgebra: the K-rank sum of A_F ’s natural module ($2 + 4 + 4 = 10$) does not equal that of the Pati–Salam natural module ($8 + 4 = 12$). The order-one carve-out from the Pati–Salam algebra to A_F is therefore a statement about the bimodule, not the algebra.*

This is what Section 6.3 of the previous document was getting at, before it stopped. The next four sections do not stop.

5 The bimodule, actually built

For one fermion generation, set

$$H_F = \mathbb{C}^{32} = \text{span} \{ |a, s, w, c\rangle : a \in \{\pm 1\}, s \in \{\pm 1\}, w \in \{\pm 1\}, c \in \{1, 2, 3, 4\} \},$$

with a indexing particle (+) vs antiparticle (−), s left (+) vs right (−), w the up/down member of the weak doublet, and c the colour, with $c \in \{1, 2, 3\}$ quark colours and $c = 4$ the lepton “colour.”

Grading and real structure. Define

$$\gamma |a, s, w, c\rangle = (a \cdot s) |a, s, w, c\rangle, \quad J |a, s, w, c\rangle = |-a, s, w, c\rangle \text{ followed by complex conjugation of coefficients}$$

With these choices: $\gamma^2 = +1$, $J^2 = +1$, $J\gamma + \gamma J = 0$. KO-dimension 6 holds on the nose. The dimension count matches: $\text{tr } \gamma = 0$, i.e. equal numbers of + and − chirality states.

Algebra action. For $(\lambda, q, m) \in A_F$, define $\pi_F(\lambda, q, m) \in \text{End}(H_F)$ as follows. On the particle sector ($a = +1$):

- Left-handed sector ($s = +1$): the quaternion q acts on the ($w = +1, w = -1$) doublet via its $M_2(\mathbb{C})$ realisation $q \mapsto \begin{pmatrix} q_0 + iq_1 & q_2 + iq_3 \\ -q_2 + iq_3 & q_0 - iq_1 \end{pmatrix}$, with c trivial.

- Right-handed up sector ($s = -1, w = +1$): scalar multiplication by λ .
- Right-handed down sector ($s = -1, w = -1$): scalar multiplication by $\bar{\lambda}$.

On the antiparticle sector ($a = -1$): $m \in M_3(\mathbb{C})$ acts on the quark colour index $c \in \{1, 2, 3\}$, and $c = 4$ is fixed (identity action). The opposite action is $\pi_F^{\text{op}}(a) = J \pi_F(a)^* J^{-1}$.

Verification. All of the following were checked at machine precision in this session. We record the worst-case residuals over the random samples or basis elements used, in double-precision arithmetic.

condition	meaning	residual
$\pi_F(1) = I$	π_F unital	0
$\pi_F(ab) = \pi_F(a)\pi_F(b)$	π_F multiplicative	1.83×10^{-15}
$\pi_F(a^*) = \pi_F(a)^*$	$*$ -preserving	0
$[\gamma, \pi_F(a)] = 0$	algebra preserves grading	0
$[\pi_F(a), \pi_F^{\text{op}}(b)] = 0$	bimodule property	0
$J^2 = +I$	KO-dim 6 sign of J	0
$J\gamma + \gamma J = 0$	KO-dim 6 anti-commutation	0
$\gamma^2 = I$	$\mathbb{Z}/2$ grading	0
$D = D^*$	Dirac is Hermitian	0
$D\gamma + \gamma D = 0$	Dirac is odd	0
$DJ = JD$	Dirac commutes with J	0
$[[D, \pi_F(a)], \pi_F^{\text{op}}(b)] = 0$	order-one for A_F	0

The non-zero residual 1.83×10^{-15} for multiplicativity is the unavoidable round-off in composing 32×32 complex matrix products in double precision; one cannot expect better. The other entries are exact zeros because the corresponding conditions are computed entry-by-entry from permutations and additions, not from matrix multiplications. In short: the linear algebra was done correctly.

The Dirac operator. Define $D \in \text{End}(H_F)$ on basis vectors by, for each (w, type) where $\text{type} \in \{\text{quark}, \text{lepton}\}$, choosing a Yukawa value $y_{w, \text{type}}$ and setting

$$D | +1, +1, w, c \rangle = y_{w, \text{type}} | +1, -1, w, c \rangle, \quad D | +1, -1, w, c \rangle = \overline{y_{w, \text{type}}} | +1, +1, w, c \rangle,$$

with the antiparticle entries fixed by $DJ = JD$. The construction makes D Hermitian, anti-commuting with γ , and commuting with J .

Order-one. For this D and for A_F :

$$[[D, \pi_F(a)], \pi_F^{\text{op}}(b)] = 0 \quad \text{for all } a, b \in A_F,$$

verified at machine precision over random samples.

The carve-out, on the nose. The natural representation of $M_2(\mathbb{H}) \oplus M_4(\mathbb{C})$ acting on the same H_F extends π_F in the canonical way: $M_2(\mathbb{H})$ acts on the (s, w) block of particles (treating (L_u, L_d, R_u, R_d) as $\mathbb{H}^2$), and $M_4(\mathbb{C})$ acts on the colour index of antiparticles including $c = 4$. Two things are then computationally manifest.

- The off-diagonal quaternionic blocks of $M_2(\mathbb{H})$ mix $s = +1$ with $s = -1$, so $[\gamma, \pi_{PS}(X)] \neq 0$ for generic $X \in M_2(\mathbb{H}) \oplus M_4(\mathbb{C})$. Residual under random sampling: ~ 8.2 , which is to say not zero.
- Restricting to the γ -preserving subalgebra $\mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$, the order-one condition for the SM Dirac fails: residual $\sim 9.6 \times 10^{-2}$.

Observation 5. *With the bimodule above, A_F satisfies the order-one condition for the SM Yukawa Dirac. The natural γ -preserving extension $\mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$ does not. The non- γ -preserving extension $M_2(\mathbb{H}) \oplus M_4(\mathbb{C})$ does not even satisfy $[\gamma, \pi] = 0$. The carve-out is, in this concrete computational realisation, a fact about the bimodule.*

This is the calculation that the previous document said one would have to do. We have now done it. The conclusion is the standard one. Section 6.3 of the previous document is, with this section, closed in the modest sense of having the calculation it pointed at on the table.

6 The Dirac-dimension functional

Once the bimodule is built, one can ask a question that the previous document did not ask, because it had not yet built the bimodule: for a subalgebra $\mathcal{A} \subseteq \text{End}(H_F)$ acting on H_F compatibly with γ and J , how large is the space of admissible Dirac operators?

Concretely, define

$$\mathcal{D}(\mathcal{A}) := \left\{ D \in \text{End}(H_F) : D = D^*, D\gamma + \gamma D = 0, DJ = JD, [[D, \pi(a)], \pi^{\text{op}}(b)] = 0 \forall a, b \in \mathcal{A} \right\},$$

a real linear subspace of $\text{End}(H_F)$.

The first three conditions (Hermiticity, anti-commutation with γ , commutation with J) carve a base subspace of $\text{End}(H_F)$. With $\dim_{\mathbb{R}} \text{End}(H_F) = 2 \cdot 32^2 = 2048$ and the conditions linearly independent, direct computation gives the base null space dimension $d_{\text{base}} = 272$. The order-one condition then cuts this 272-dimensional space further, by an amount depending on $\mathcal{A}$.

We computed $\dim_{\mathbb{R}} \mathcal{D}(\mathcal{A})$ for a chain of subalgebras. Setting up the constraints in a basis of the base null space and taking ranks via QR with column pivoting, the result is the following.

Algebra $\mathcal{A}$	$\dim_{\mathbb{R}} \mathcal{A}$	$\dim_{\mathbb{R}} \mathcal{D}(\mathcal{A})$
none (i.e. only the unit)	1	272
$\mathbb{C}$	3	146
$\mathbb{H}$	5	146
$\mathbb{C} \oplus \mathbb{H}$	7	80
$M_3(\mathbb{C})$	19	128
$\mathbb{C} \oplus M_3(\mathbb{C})$	21	16
$\mathbb{H} \oplus M_3(\mathbb{C})$	23	16
$A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$	25	16
$\mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$	41	8

Two things are worth noting before any interpretation.

Observation 6. $\dim \mathcal{D}$ is not a monotone function of $\dim \mathcal{A}$. $M_3(\mathbb{C})$ (dim 19) gives $\dim \mathcal{D} = 128$, while the strictly smaller $\mathbb{C} \oplus \mathbb{H}$ (dim 7) gives $\dim \mathcal{D} = 80$. The functional depends on which directions of $\text{End}(H_F)$ the algebra hits, not on the algebra's dimension.

Observation 7. Three different subalgebras of A_F give $\dim \mathcal{D} = 16$: the algebras $\mathbb{C} \oplus M_3(\mathbb{C})$, $\mathbb{H} \oplus M_3(\mathbb{C})$, and A_F itself. The full algebra A_F does not impose any further constraint on $\mathcal{D}$ beyond what either $\mathbb{C} \oplus M_3(\mathbb{C})$ or $\mathbb{H} \oplus M_3(\mathbb{C})$ already imposes.

The number 16 is not arbitrary. Per generation, the SM Yukawa parameters are encoded as a 2×2 Yukawa matrix per fermion type (quark, lepton), each entry a complex number. Modulo Hermiticity and J -pairing, this gives $2 \times 4 \times 2 = 16$ real parameters per generation. The plateau at 16 is exactly the per-generation SM Yukawa parameter count.

This is checkable. Set up 16 explicit physical Yukawa basis Diracs, one per (type, w_L , w_R , re/im). Each one was verified to lie in $\mathcal{D}(A_F)$ with residual $\leq 1.6 \times 10^{-14}$. Their image in the 16-dimensional coordinate space of $\mathcal{D}(A_F)$ has rank 16. So:

Theorem 8. $\mathcal{D}(A_F)$ has dimension 16, and is spanned exactly by the 16-parameter family of physical Yukawa Diracs (per generation, modulo Hermiticity and J -pairing).

This is what Chamseddine–Connes–van Suijlekom 2013 say. With the bimodule on the table, we have just checked it.

7 A_F is not picked out by $\dim \mathcal{D}$ alone

Observation 7 says the Dirac-dimension functional, taken by itself, does not single out A_F among its subalgebras. We pushed on this and verified the strongest version of the statement.

Theorem 9. As subspaces of $\text{End}(H_F)$,

$$\mathcal{D}(\mathbb{C} \oplus M_3(\mathbb{C})) = \mathcal{D}(\mathbb{H} \oplus M_3(\mathbb{C})) = \mathcal{D}(A_F).$$

Proof. The first two subalgebras are contained in A_F , so each imposes a subset of A_F 's constraints, hence each $\mathcal{D}$ contains $\mathcal{D}(A_F)$. By Theorem 8 and Observation 7 all three spaces are 16-dimensional. Containment of finite-dimensional spaces of equal dimension is equality.

Numerical confirmation: we computed orthonormal bases for all three subspaces and projected one onto another. The projection residuals were $\leq 7.2 \times 10^{-15}$ in all three pairwise comparisons. $\square$

This sharpens, in a way that is initially surprising and on reflection slightly embarrassing, what the Dirac-dimension functional can and cannot do.

Observation 10. The $\mathbb{H}$ summand of A_F imposes zero additional order-one constraint on $\mathcal{D}$ beyond what is already imposed by $\mathbb{C} \oplus M_3(\mathbb{C})$. Symmetrically, the $\mathbb{C}$ summand imposes zero additional constraint beyond what $\mathbb{H} \oplus M_3(\mathbb{C})$ already imposes. The two summands $\mathbb{C}$ and $\mathbb{H}$ each independently suffice to cut $\mathcal{D}(M_3(\mathbb{C}))$ (dim 128) down to the SM Yukawa subspace (dim 16). Adding the other contributes nothing further at the level of $\mathcal{D}$.

This is a real structural fact. It also means that any honest variational attempt to pick A_F out using $\dim \mathcal{D}$ as the functional will fail. The maximum is attained on a plateau, and the plateau contains A_F as one point among at least three. Any unique selection of A_F must use additional information.

The natural additional information is the gauge group $U(\mathcal{A})$. We record the relevant dimensions and note the salient gap.

- $\dim U(\mathbb{C} \oplus M_3(\mathbb{C})) = 1 + 9 = 10$, with $U(\mathbb{C} \oplus M_3(\mathbb{C})) = U(1) \times U(3)$. No $SU(2)$ factor.
- $\dim U(\mathbb{H} \oplus M_3(\mathbb{C})) = 3 + 9 = 12$, with $U(\mathbb{H} \oplus M_3(\mathbb{C})) = SU(2) \times U(3)$.
- $\dim U(A_F) = 1 + 3 + 9 = 13$, with $U(A_F) = U(1) \times SU(2) \times U(3)$, reducing modulo $\mathbb{Z}_6$ to the SM gauge group.

The two non- A_F algebras at $\dim \mathcal{D} = 16$ are inadequate for the following different reasons.

- $\mathbb{C} \oplus M_3(\mathbb{C})$: no $SU(2)_L$. No W bosons.
- $\mathbb{H} \oplus M_3(\mathbb{C})$: has $SU(2)$, has $U(1)$, but the $U(1)$ is the determinant phase of $U(3)$ and the $\lambda \leftrightarrow \bar{\lambda}$ distinction on right-handed up vs. down fermions is absent. The Standard Model hypercharge assignments $Y(u_R) \neq Y(d_R)$ require the action of the $\mathbb{C}$ summand on the bimodule, where one gets λ on u_R and $\bar{\lambda}$ on d_R . Without that summand the hypercharge assignments are wrong.

Putting these together:

Observation 11. A_F is the minimal $*$ -subalgebra of the natural γ -preserving extension of the SM bimodule such that

- $\dim_{\mathbb{R}} \mathcal{D}(\mathcal{A}) \geq 16$ (full SM Yukawa freedom), and
- the unitary group $U(\mathcal{A})$ contains both the $SU(2)_L$ factor and the $U(1)$ factor that distinguishes $Y(u_R)$ from $Y(d_R)$ in the bimodule action.

Two conditions. Not one.

8 The 8-dimensional Pati–Salam Dirac space

The carve-out from Pati–Salam to A_F , then, is from $\dim \mathcal{D} = 8$ to $\dim \mathcal{D} = 16$. The factor of two suggests a clean structural identification, and indeed there is one.

For the algebra $\mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$ (which preserves γ) acting on the same bimodule via the canonical extension of π_F , we computed $\mathcal{D}$ as an 8-dimensional subspace of $\mathcal{D}(A_F)$, and expressed each of its basis vectors in the physical-Yukawa coordinate system of Theorem 8.

The output is, in every case, of the following form. A basis vector of the 8-dimensional surviving Dirac space, when written in physical Yukawa coordinates, has the structure

$$\sum_j c_j \cdot (\text{quark Yukawa})_j + \sum_j c_j \cdot (\text{lepton Yukawa})_j,$$

with the same coefficient c_j on the quark term and on the corresponding lepton term. Each of the 16 individual Yukawa basis directions (8 quark, 8 lepton) violates the additional constraints by an $O(1)$ amount. Only quark–lepton paired combinations survive.

Theorem 12. *The eight surviving Dirac directions in $\mathcal{D}(\mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C}))$ are exactly the leptoquark-unified combinations: a combined (quark+lepton) Yukawa for each of the 8 ($w_L, w_R, \text{re/im}$) assignments, with equal weight on quark and lepton.*

This is, in plain language, the Pati–Salam prediction that leptons are the fourth colour, hence that lepton and quark Yukawas are tied at the unification scale. Numerically, after RG running, this is the source of the $m_b/m_\tau \approx 3$ low-scale prediction. The carve-out from $\mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$ to A_F is the breaking of this tie: the order-one condition for the $\mathbb{C}$ summand of A_F acting only on right-handed Standard Model particles, together with the restriction from $M_4(\mathbb{C})$ to $M_3(\mathbb{C}) \oplus$ (trivial on leptons), is precisely what allows independent quark and lepton Yukawas.

We did not, in this session, observe anything in this analysis that is not already in Chamseddine–Connes–van Suijlekom 2013. What is new is the computational receipt: the 8-dimensional space is, as a subspace of the 16-dimensional one, exactly the diagonal quark–lepton matched subspace. We have not seen this stated in this form in the published literature. We have also not searched the published literature exhaustively. So either this is a known consequence stated in slightly different language elsewhere, or it has been computed before without being highlighted. Both possibilities are likely; both possibilities are fine; the receipts are in `phase6_8dim.py`.

9 Step 4, restated

The previous document stated the open question this way:

The variational principle that would close the framework, if it exists, is a functional on $\mathcal{M} = \{\text{Krajewski diagrams of KO-dim 6}\} \times \{\text{compatible Dirac moduli}\}$ whose minimum (or canonical critical point) is the SM bimodule.

This is still the question. We can now say slightly more about what such a functional must do.

Observation 13. *A functional F on $\mathcal{M}$ that singles out A_F uniquely cannot factor through $\dim \mathcal{D}$ alone, because that functional plateaus. It must combine, in some genuine way, at least the following two kinds of information:*

- *information about the bimodule structure (KO-dimension, order-one carve-out, dimension of admissible Dirac operators);*
- *information about the unitary group of $\mathcal{A}$ acting on the bimodule (the gauge content, including hypercharge assignments).*

A merely topological or merely representation-theoretic functional will not suffice; the function must see both kinds of structure simultaneously. The candidate spectral-action functional has this property in principle, since the heat-kernel coefficients a_2, a_4 involve both eigenvalue content of D and representation content of $\mathcal{A}$. Whether the unique minimizer is A_F remains, as of May 2026, open.

The previous document’s framing of Step 4 stands. We add one observation: the naive simplification “maximize the dimension of the admissible Dirac space” does not work, and the reason it does not work is now visible at the level of explicit subspaces. A successful single functional must do more than count.

10 Residues, after this round

The residues from the previous document remain residues. The only one we touched is Koide, and its status is unchanged in substance.

Residue (Koide, current status.). $K = 0.6666645(5)$ in the pole-mass scheme using PDG 2024 data; consistent with $2/3$ to within 0.43σ ; not preserved under full SM running. A tau-mass

shift of +0.04 to +0.11 MeV (inside current uncertainty) would make it exact. No derivation; no quark extension. Coincidence or hint, currently undecidable. Per Observation 2, the seven-decimal version of the statement is scheme-specific.

Residue (Three generations.). *No change. Empirical fact, no principled derivation in any rigorous framework.*

Residue (Cosmological constant.). *No change. Worst quantitative discrepancy in theoretical physics; addressed by no current framework.*

Residue (Selection of A_F). *Sharper after Sections 6–9. Not closed. The cleanest available characterisation of A_F now goes: it is the minimal $*$ -subalgebra such that (i) the order-one condition on the SM bimodule admits the full 16-dimensional Yukawa space, and (ii) the unitary group action carries the SM gauge group with correct hypercharge assignments. This is a two-condition characterisation, not a one-functional minimisation. Closing Step 4 still requires the latter.*

Residue (Unreasonable effectiveness of operator algebras.). *No change. The alignment between purely internal operator-algebraic structures and physical observers, measurements, scales, and effective theories remains unexplained. We did not propose an explanation.*

Residue (The meta-puzzle, after Wigner.). *No change. Still 1960.*

11 Verdict, updated

The previous document’s verdict stands without subtraction. We add three lines.

The Generator hypothesis, as originally proposed, did not survive honest scrutiny. That has not changed.

The honest rewrite (categorical RG on operator algebras) is mathematical physics in a particular dialect and is not new. That has not changed.

The closest existing program (Connes’s spectral action) recovers the bosonic Standard Model from one functional given A_F . That has not changed.

What changed in this session:

- The Section 6.3 calculation that the previous document gestured at is now on the table, end-to-end, at machine precision.
- One parenthetical sentence of the previous document (Observation 4) was corrected.
- The functional approach to Step 4 was tested at the level of $\dim \mathcal{D}$, and shown to plateau in a way that rules out the naive single-functional minimisation. The successful functional must see both bimodule structure and gauge content, simultaneously.
- The structural meaning of the 8-vs-16 dimensional carve-out was located: it is precisely the breaking of Pati–Salam leptoquark unification at the level of Yukawa parameters.

None of these closes Step 4. The previous document’s summary remains:

We did not close it. We did locate it.

We located it slightly more precisely. That is the substantive content of this document, and it is smaller than the calculation that produced it would suggest. This is normal. Honest work has this shape.

12 What this document is

This document is, like its predecessor, a record. It is not a theory, not a framework, not a derivation of the universe from one equation. The arithmetic in it was done in Python in a sandboxed environment with a wall clock; the linear algebra ran in BLAS; the SVDs were computed with LAPACK; the constructions matched the published literature wherever they overlapped. Receipts for everything in Sections 4–8 are in the files `phase2a_explicit_embedding.py`, `phase2c_carveout.py`, `phase3_dirac_dim_v2.py`, `phase4_fast.py`, `phase5_structure.py`, and `phase6_8dim.py`. The numerical precision of every reported residual can be reproduced by anyone with a Python installation; the patience required is approximately one hour of compute on a 4 GB single-core machine.

The strange idea, in the form it survives after this further round, is exactly as strange as it was before, and not strange in any new way. It still points at one open question. It has not become a theory; it has not become a framework. The thing it did, in this round, was to acquire one more layer of precision around the open question. The next round, if there is one, would either close the question or determine that it cannot be closed without genuinely new mathematical ideas.

Both possibilities remain interesting. Neither is currently known. The strangeness has not gone anywhere. It has been measured a little more carefully.

That is enough.

A Computational receipts

This section records, for each substantive computational claim in the body of the document, the file in which it was carried out, the principal verification performed, and the worst-case numerical residual obtained. All computations were performed in Python with `numpy` 2.4.4 and `scipy` 1.17.1 in double precision, in a sandboxed environment with no external network access. The wall-clock cost of reproducing the full set is approximately one hour on a single-core 4 GB machine.

file	verifies	worst residual
phase1_koide.py	Koide ratio under CODATA 2022 and PDG 2024; tau-mass shift for $K = 2/3$; σ -distance	data-driven (see Sec. 3)
phase2a_v2.py	enumeration of 9421 finite real semisimple $*$ -algebras up to real dim 96 and 5 summands; CC07-admissible count 91; unital $*$ -subalgebra obstruction for A_F in fermion-dim-32 candidates	combinatorial
phase2a_explicit_embedding.py	the explicit map $\iota : A_F \rightarrow M_2(\mathbb{H}) \oplus M_4(\mathbb{C})$ of Theorem 3; unitality, multiplicativity, additivity, $\mathbb{R}$ -linearity, $*$ -preservation, faithfulness over 50 random elements each	1.78×10^{-15}
phase2c_carveout.py	the full bimodule of Section 5; all entries of the verification table; order-one for A_F ; the $[\gamma, \pi_{PS}] \neq 0$ obstruction; order-one failure for $\mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$	1.83×10^{-15} for non-zero entries; obstruction residuals 8.23 and 9.63×10^{-2}
phase3_dirac_dim_v2.py	$d_{\text{base}} = 272$ via SVD of the stacked Hermitian+anti- γ +commute- J constraint matrix; $\dim \mathcal{D}(A_F) = 16$; $\dim \mathcal{D}(\mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})) = 8$ via QR-based rank determination	SVD tolerance $10^{-10} \cdot \sigma_{\max}$
phase4_fast.py	the full chain of subalgebras in Section 6 using batched <code>numpy</code> order-one constraint computation; the plateau $\dim \mathcal{D} = 16$ for $\mathbb{C} \oplus M_3$, $\mathbb{H} \oplus M_3$, and A_F ; identification of the 16 physical Yukawa basis Diracs and rank check	$\leq 1.64 \times 10^{-14}$
phase5_structure.py	subspace equality $\mathcal{D}(\mathbb{C} \oplus M_3) = \mathcal{D}(\mathbb{H} \oplus M_3) = \mathcal{D}(A_F)$ via projector residual; containment in $\mathcal{D}(M_3)$; projection of the explicit SM Yukawa Dirac onto $\mathcal{D}(A_F)$; rank verification of the physical Yukawa basis	$\leq 7.20 \times 10^{-15}$ for subspace equality; 6.69×10^{-17} for SM Dirac projection
phase6_8dim.py	the 8-dimensional Dirac space of $\mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$ computed by projecting its additional order-one constraints onto $\mathcal{D}(A_F)$; expression of each surviving basis direction in the physical Yukawa coordinate system; verification that quark and lepton coefficients in each surviving direction are paired (equal)	matched-coefficient pairing exact within tolerance 10^{-8}

The python files are not included here; they exist in the working environment in which this document was produced. Anyone wishing to reproduce the computations should be able to write equivalent scripts in less than a day from the specifications in Sections 4 through 8 of this document.

B Notes on what a successful Step 4 functional must do

We do not, in this document or elsewhere, propose a functional that closes Step 4. Observation 13 restricts the shape of any such functional. We record here a slightly longer list of properties such a functional F would need to have, in case it is useful for anyone attempting to construct one.

1. F must depend on the bimodule (A, H_F, γ, J, π) , not just on the algebra A . The Dirac-dimension functional already has this property; the plateau at $\dim \mathcal{D} = 16$ across three different subalgebras of A_F shows that bimodule structure alone is not sufficient either.
2. F must distinguish A_F from $\mathbb{H} \oplus M_3(\mathbb{C})$ despite the two having the same Dirac space. The visible discriminator is the action of the $\mathbb{C}$ summand of A_F on the right-handed up vs. down sectors via λ vs. $\bar{\lambda}$, which is what gives the $Y(u_R) \neq Y(d_R)$ structure. Any successful F must therefore see hypercharge assignments at the level of the representation, not just at the level of the gauge group.
3. F must distinguish A_F from $\mathbb{C} \oplus M_3(\mathbb{C})$ despite the same Dirac space. Here the visible discriminator is the presence of $SU(2)_L$ in the gauge group. Any successful F must therefore involve a non-abelian piece.
4. F must, taken across the chain of Section 6, single out A_F over $\mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$, i.e. choose the SM over Pati–Salam. This requires F to prefer the un-unified Yukawa structure ($\dim \mathcal{D} = 16$) over the leptiquark-unified structure ($\dim \mathcal{D} = 8$). On the face of it, this looks like preferring *more* Dirac freedom, which is a sensible criterion. Any successful F must, however, also choose A_F specifically over its other equally-Dirac-free subalgebras, per points (2) and (3) above. Hence F cannot be simply “maximise $\dim \mathcal{D}$.”
5. The spectral action functional $S[D] = \text{Tr } f(D/\Lambda) + \langle \psi, D\psi \rangle$ is the natural candidate, because the heat-kernel coefficients a_n mix Dirac spectrum content with representation content of A . The question of whether the spectral action’s value at the SM is actually a critical point in A (not just in D) does not appear to have been settled in the literature as of May 2026.
6. Khalkhali–Pagliaroli [11] propose a moment-method / SDP bootstrap on Dirac ensembles, which provides numerical evidence for clustering of Dirac spectra near specific algebras. The bootstrap as currently formulated does not select A_F uniquely either, but gives the closest existing tractable framework in which one could test a candidate F .

We list these because they constitute a kind of negative specification: any proposal for a Step 4 functional that fails any of (1)–(4) cannot be right, and any proposal that passes all of them is worth taking seriously. We have not produced such a proposal. This is, again, where the genuine open question lives.

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